INVERTIS UNIVERSITY BAREILLY
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# STUDY AND EVALUATION SCHEME
## (With effective from academic session 2023-2024)
## B. Tech. in Computer Science & Engineering with specialization in Cloud Computing
## YEAR I, SEMESTER I

| Sl. No. | Category | Course Code | Course Title/ Subjects | Hours per week | | | Evaluation Scheme | | Total | Credits |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | L | T | P | CA | EE | | |
| **THEORY** | | | | | | | | | | |
| 1 | Ability Enhancement | ENG105 | English | 2 | 1 | 0 | 25 | 50 | 75 | 3 |
| 2 | Science | MAT 102 | Engineering Mathematics-I | 3 | 1 | 0 | 30 | 70 | 100 | 4 |
| 3 | Science | PHY 105 | Physics for Engineers–I | 3 | 1 | 0 | 30 | 70 | 100 | 4 |
| 4 | Engineering | BEE 101 | Basic Electrical and Electronics | 3 | 1 | 0 | 30 | 70 | 100 | 4 |
| 5 | Engineering | BCSI101 | Problem Solving and Programming in C | 3 | 0 | 0 | 25 | 50 | 75 | 3 |
| 6 | Engineering Science Course | IIOT1 | Fundamentals of IOT Development | 4 | 0 | 0 | 30 | 70 | 100 | 4 |
| **PRACTICALS AND PROJECTS** | | | | | | | | | | |
| 6 | Science | PHY 106 | Physics for Engineers Lab–I | 0 | 0 | 2 | 10 | 15 | 25 | 1 |
| 7 | Engineering | BEE 102 | Basic Electrical and electronics lab | 0 | 0 | 2 | 10 | 15 | 25 | 1 |
| 8 | Engineering | BCSI102 | Problem solving and Programming in C lab | 0 | 0 | 2 | 10 | 15 | 25 | 1 |
| 9 | Engineering | BME101 | Workshop practice | 0 | 0 | 2 | 10 | 15 | 25 | 1 |
| 10 | Engineering | BCSI103 | IT Workshop | 0 | 0 | 2 | 10 | 15 | 25 | 1 |
| 11 | Skill Enhancement | ENG 113 | Soft Sills and Self Awareness | 0 | 0 | 2 | 10 | 15 | 25 | 1 |
| | | | **Total** | **18** | **4** | **12** | **230** | **470** | **750** | **28** |

**L**-Lecture, **T**- Tutorial , **P**- Practical ,**CA**- Continuous Assessment, **EE**- End Semester Examination

[Revised after the inclusion of Skill and Entrepreneurship Courses effective from the session 2022-23]

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# STUDY AND EVALUATION SCHEME
## (With effective from academic session 2023-2024)
## B. Tech. in Computer Science & Engineering
## with specialization in Cloud Computing
## YEAR I, SEMESTER II

| Sl. No. | Category | Course Code | Course Title/ Subjects | Hours per week | | | Evaluation Scheme | | Total | Credits |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | L | T | P | CA | EE | | |
| **THEORY** | | | | | | | | | | |
| 1 | Ability Enhancement | ENG106 | Professional English | 2 | 1 | 0 | 25 | 50 | 75 | 3 |
| 2 | Science | MAT103 | Engineering Mathematics–II | 3 | 1 | 0 | 30 | 70 | 100 | 4 |
| 3 | Science | PHY107 | Physics for Engineers–II | 3 | 1 | 0 | 30 | 70 | 100 | 4 |
| 4 | Science | CHY103 | Environmental Science | 2 | 0 | 0 | 15 | 35 | 50 | 2 |
| 5 | Engineering | BCSI201 | Design Thinking | 3 | 0 | 0 | 25 | 50 | 75 | 3 |
| 6 | Engineering | BCSI202 | Web Designing | 3 | 0 | 0 | 25 | 50 | 75 | 3 |
| 7 | Engineering Science Course | IIOT2 | Industrial Communication Protocols & Connectivity | 4 | 0 | 0 | 30 | 70 | 100 | 4 |
| **PRACTICALS AND PROJECTS** | | | | | | | | | | |
| 7 | Ability Enhancement | ENG107 | Communication Techniques lab | 0 | 0 | 2 | 10 | 15 | 25 | 1 |
| 8 | Science | PHY108 | Physics for Engineers lab –II | 0 | 0 | 2 | 10 | 15 | 25 | 1 |
| 9 | Engineering | BCSI203 | Web Designing lab | 0 | 0 | 2 | 10 | 15 | 25 | 1 |
| 10 | Engineering | BCSI204 | Computer Aided Graphics | 0 | 0 | 2 | 10 | 15 | 25 | 1 |
| 11 | Skill Enhancement | ENG114 | Soft Skills and Communication | 0 | 0 | 2 | 10 | 15 | 25 | 1 |
| 12 | Skill Enhancement | SEP200 | Extra-curricular Activity ( NSS/NCC/Scouting/ Club Activity) | 0 | 0 | 2 | 10 | 15 | 25 | 1 |
| | | | **TOTAL** | 20 | 3 | 12 | 240 | 485 | 725 | 29 |

**L-**Lecture, **T-** Tutorial , **P-** Practical ,**CA-** Continuous Assessment, **EE-** End Semester Examination

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# ENG 105: ENGLISH

**Unit 1: Prose-** A Cup of Tea by Katherine Mansfield, The Piece of String by Guy De Maupassant, Text of Steve Jobs' Commencement address -2005.
**Poems:** Ode to the skylark – by P B Shelley, Where the Mind Is Without Fear – by Rabindranath Tagore, The Road Not Taken- Robert Frost.

**Unit 2**: **Grammar & Vocabulary-** Tenses and the concept of time, Verb Types, Active and Passive Voice, Narration, Prepositions, Conditionals, Modal Auxiliaries, Conjunctions, One-word substitutions, Synonyms and Antonyms.

**Unit 3: Writing Skills-** Writing Paragraph, Essay, Writing Articles for Newspapers & Magazines, Writing Applications, Resumé Writing.

**Unit 4: Reading Skills-** Reading Comprehension, Summarizing and Note making.

**Unit 5: Speaking Skills-** Conversations- Introduction, Purpose, Features, Delivering Speeches- Welcome, Introductory, Vote of Thanks, Farewell Speech, Indianism.

**Text/Reference Books**


1. Communication Skills, Pushp Lata & Sanjay Kumar, Oxford Higher Education/Oxford University Press.
2. Technical Communication; Principles and Practice, Meenakshi Raman & Sangita Sharma, Oxford University Press.
3. Effective Technical Communication, M Ashraf Rizvi, Tata Mcgraw Hill.

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# MAT 102: ENGINEERING MATHEMATICS-I

**Unit 1: Differential Equation I-** Differential equations of first order and first degree - Linear differential equations, Reducible to linear form, Exact Form, Reducible to Exact Form, Linear Differential Equations of Higher Order with Constant Coefficients Only.

**Unit 2: Differential Equation II-** Second order ordinary differential equations with variable coefficients-Homogeneous form, Exact form, Change of Dependent Variable, Change of Independent Variable, Normal form, Variation of Parameters.

**Unit 3: Differential Calculus I-** Partial Differentiation, Euler's Theorem on Homogeneous Functions, Approximate Calculations, Maxima and Minima of two and more independent variables, Lagrange's Method of Multipliers.

**Unit 4: Differential Calculus II-** Asymptotes (Cartesian co-ordinates only), Curvature, Concavity, Convexity and Point of Inflexion (Cartesian co-ordinates only), Curve Tracing (Cartesian and Standard Polar Curves).

**Unit 5: Integral Calculus-** Area of Curves, Rectification, Surface and Volume of Solids of Revolution, Double and Triple Integrals, Double Integral by changing into polar form, Area and Volume by Double Integration, Change of Order of Integration, Beta Function and Gamma Function (Simple Properties).

**Text/Reference Books**


1. Higher Engineering Mathematics, B.V. Ramana, Tata McGraw Hill.
2. Advanced Engineering Mathematics, Erwin Kreyszig, Wiley 9$^{th}$ Edition
3. Calculus and Analytical Geometry, Thomas and Finney, Narosa Publishing House(New Delhi)
4. Integral Calculus, Shanti Narayan, S. Chand.
5. Differential Calculus, Shanti Narayan, S. Chand.
6. Ordinary and Partial differential equations, M. D. Raisinghania, S. Chand.
7. Calculus, Thomas & Finney, Narosa Publishing House(New Delhi).

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# PHY 105: PHYSICS FOR ENGINEERS –I

**Unit I: Basic Optics:** Reflection of light, Refraction of light, total internal reflection and its applications, Refraction and dispersion of light through a prism. Wave front and Huygens's principle, reflection and refraction of plane wave at a plane surface using wave fronts. Young's double slit experiment and expression for fringe width, coherent sources. Optical fiber as optical wave-guide. Numerical aperture and maximum angle of acceptance.

**Unit II: Interference of Light:** Michelson's Interferometer: Production of circular & straight-line fringes, Determination of wavelength of light. Determination of wavelength. Newton's rings and measurement of wavelength of light. Elementary idea of anti-reflection coating.

**Unit III: Polarization of Light:** Plane circular and elliptically polarized light on the basis of electric (light) vector, Malus law. Double Refraction: Qualitative description of double refraction phase retardation plates, quarter and half wave plates, construction, working and use of these in production and detection of circular and elliptically polarized light. Optical Activity: Optical activity and law of optical rotation, specific rotation and its measurement using the bi-quartz device.

**Unit IV: Diffraction of Light:** Single slit diffraction: Quantitative description of single slit, position of maxima / minima and width of central maximum, intensity variation. Diffraction Grating: Construction and theory. Determination of wavelength of light using plane transmission grating. Resolving power: Geometrical & Spectral, Raleigh criterion, Resolving power of diffraction grating.

**Unit V: Coherence & Laser:** Spatial and temporal coherence, Coherence length, Coherence time and 'Q' factor for light. Temporal coherence and spectral purity, Theory of laser action: Einstein's coefficients, Components of a laser. Theory, Design and applications of He-Ne and semiconductor lasers.

**Text/Reference Books**


1. Optics, Ajay Ghatak, Tata McGraw Hill(Noida).
2. Optics, N. Subrahnmanyam & Brij Lal, S. Chand (New Delhi).
3. Feynman Lectures on Physics, Volume I, II & III, Perseus Books Group California Institute of Technology.
4. Fundamental of optics, white & jenkins , Tata McGraw Hil.
5. Optics, Eugene Heceht, Schaum series.

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# BEE 101: BASIC ELECTRICAL AND ELECTRONICS

**Unit 1: Basic Electrical Quantities:** -Basic concept of charge, current, voltage, resistance, power, energy and their units, Conversion of units of work, power and energy from one form to another. Electrical Energy: Difference between ac and dc and their applications, Classification of two terminal elements, Energy Sources, Source Transformation, ideal independent two terminal electrical sources.

**DC Networks**: Ohm's law, resistances, capacitances & inductors in series and parallel, Kirchhoff's laws and their applications in solving electrical network problems, Node Voltage and Mesh Current Analysis, Star-delta transformation.

**Unit 2: Network theorems:** Thevenin Theorem, Norton Theorem and Superposition Theorem, Statement and applications including dependent sources.

**Transformer:** Faraday's Law of Electromagnetic Induction, Construction and Operation of Single-Phase Transformer, EMF Equation, Voltage & Current Relationship and Phasor Diagram of Ideal Transformer.

**Unit 3: AC Fundamentals:** Concept of alternating current and voltage, Generation of Single -Phase AC Voltage, EMF Equation, Average, RMS and Effective Values Representation of alternating sinusoidal quantities by vectors, Phasor algebra (addition, subtraction, multiplication and division of complex quantities). RL, RC & RLC Series-Parallel Circuits, Complex Representation of Impedances, Reactance, Phasor Diagram, Power and Power Factor.

**Unit 4: Review of Semiconductors:** Semiconductors, conductors and insulators, intrinsic and extrinsic semiconductors and conduction in semiconductors**.**

**Semiconductor Physics:** Basic material properties of semi-conductors, governing factors for Fermi-level, carrier concentration and carrier mobility, recombination and carrier lifetime, carrier drift, diffusion, Hall effect and continuity equation.

**Unit 5: Junction Diodes:** Introduction, forward and reverse biasing of diode, voltage current characteristic of diode, diode models, half wave rectifier, full wave rectifier, Zener diode and its application.

**Bipolar Transistor:** Transistor structure, basic transistor operation, common base configuration and its characteristics, transistor amplifying action, common emitter configuration and its characteristics, common collector configuration, limit of operation, the dc operating point and biasing techniques (fixed bias, voltage divide bias, voltage feedback type, current feedback type, and combination of voltage and current feedback types)

## Text/Reference Books


1. Integrated Electronics Analog and Digital Circuits & Systems, J. Millman & C.C. Halkias, McGraw Hill.
2. Electronic Devices and Integrated Circuits, B. P. Singh & Rekha Singh, Pearson Education.
3. Theory and problems of Basic Electrical Engineering, D.P. Kothari & I. J. Nagrath, PHI.
4. Basic Electrical Engineering, V.N. Mittal & A. Mittal, TMH.
5. Solutions of problems in Electrical Engineering, Smith Parker, CBS Publisher.
6. Electronics Devices & Circuit Theory, Boylestad & Nashelskey, Prentice Hall of India.

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# BCSI101: Problem Solving and Programming in C

**Unit 1: Introduction:** -What is a program? What is a programming language? Steps in Programming, Skills needed to do programming, A little introduction to C, writing a Program, Fundamentals of a Programming Language, Different Programming Techniques, Procedural Programming, Modular Programming, Object Oriented Programming, Getting started with compiler**.** Words and Sentences in C Language: Alphabets in C, Keywords in C, Rules of forming Words in C language, Data Variables, Data Types and Rules for naming and declaring data, variables, Basic Data Types in C, Constants, Comments in C.

**Unit 2: Instructions and Rules for Writing:** -Types of instructions, Data Manipulation Instructions, Input/output Instructions, Flow Control Instructions: Decision Control Instructions, If, if-else, If-else-if, Nested if-else, Loop Control Instructions, For Loop, While Loop, Do While, Selection Instructions.

**Unit 3: Functions:** -Why use Functions, Components of Function, Name of a function, Body of a function, Local variables of a function, Parameters or Arguments to a function, Return Values, Prototype of a function. **Arrays:** What is an array? Array Declaration, Array Initialization, accessing individual elements of an array, Two Dimensional Arrays, passing an array element to a function, Rules of using an array.

**Unit 4: Pointers:** -What is a pointer? Declaring a Pointer variable, initializing a pointer variable, Using a Pointer Variable, Pointer Arithmetic, Pointers and array, passing an entire array to a function.
Strings: What are strings? String I/O, String Manipulation Functions.

**Unit 5: Structures:** -Declaring and Accessing Structure, variables Uses of Structures, Unions Storage Classes and Scoping: Automatic, Register, External, Static, Scope of a Variable File Input/Output: Command-line arguments, File Input and Output, Combining Command-line Arguments and File I/O.

**Text/Reference Books**


1. The C Programming Language, Brian W. Kernighan, Dennis M. Ritchie, Prentice-Hall (New Delhi).
2. C Programming: A Modern Approach, K. N. King, Prentice-Hall (New Delhi).
3. C Primer Plus, Stephen Prata, Sams.
4. Practical C Programming, Steve Oualline, O'Reilly Media.
5. Let us C, Yashwant Kanetkar, BPB Publications.
6. Pointers in C, Yashwant Kanetkar, BPB Publications.

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# BME 101: WORKSHOP PRACTICE

**Carpentry Shop**

1. T – Lap joint
2. Cross lap joint

**Welding Shop**

1. Gas welding practice by students on mild steel flat
2. Lap joint by gas welding
3. Arc welding practice by students
4. Square butt joint by Arc welding

1. Job on lathe with one step turning and chamfering operations
2. Job on lathe with Facing and knurling operations
3. Drilling two holes of size 5 and 12 mm diameter on job used / to be used for shaping
4. Grinding a corner of above job on bench grinder

1. Finishing of two sides of a square piece by filing
2. To cut a square notch using hacksaw and to drill three holes on PCD and tapping

**Text/Reference Books**


1. Workshop Practice, K. C. JOHN, PHI Learning Pvt. Ltd.
2. Modern Workshop Technology, Henry Wright, BAKER Cleaver-Hume Press.
3. Workshop Technology, Hajra, Chaudhary, Media promotors & publishers pvt. Ltd.
4. Workshop Technology, B.S. Raghhuwanshi, Dhanpat Rai & Company (p) Limited.
5. Workshop Technology, Virendra Narula, Kataria & sons publications.
6. Comprehensive Workshop Technology, S.K. Garg, Laxmi Publications.

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# ENG 106: PROFESSIONAL ENGLISH

**Unit 1: Fundamentals of Communication-** Introduction, Definition, Process, Importance, Different Forms and Purpose of Communication, Barriers to Communication, Organizational and Interpersonal Communication

**Unit 2: Group Discussion-** Introduction to Group Discussion, Types, Roles and Functions in Group Discussion, Difference between GD and Debate, Preparation Strategy, Tips for a good GD.

**Unit 3: Presentation-** Fundamentals of Presentation, Audience Analysis, Organizing Material, Visual Aids and Nuances of Delivery, Body language and Effective Presentation, Question- Answer Session

**Unit 4: Professional Writing-** Official Correspondence – Drafting E- mails, Memorandum, Notice, Agenda, Minutes, Circulars, Business Correspondence-Business letter writing- Sales letters, Enquiry letters and replies to enquiry (enquiry about a product, service or information, asking for a quotation, placing an order and replies to the same) letters of Claim and Adjustment

**Unit 5: Technical Writing-** Report Writing- General and Technical report, Definition, Types, structure, Technical proposals- Definitions, Types and Format

**Text/Reference Books**


1. Communication Skills**,** Pushp Lata & Sanjay Kumar.Oxford Higher Education/Oxford University Press.
2. Technical Communication; Principles and Practice, Meenakshi Raman & Sangita Sharma,Oxford University Press.
3. Effective Technical Communication, M Ashraf Rizvi, Tata McGraw-Hill Education
4. Basic Communication Skills for Technology, Andrea J. Rutherford, Peerson Education Asia

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# MAT 103: ENGINEERING MATHEMATICS II

**Unit 1: Matrices-** Rank of a matrix, Rank of matrix by reducing to normal forms, Consistency and redundancy of systems of simultaneous linear equations and its solution, Eigen values and Eigen vectors, Cayley- Hamilton theorem (without proof), Diagonalization of matrix.

**Unit 2: Differential Equations-** Series Solutions of Second Order Linear Differential Equations with Variable Coefficients (complementary functions only), Partial Differential Equations of First Order: Lagrange's Form, Standard Forms, Charpit's Method.

**Unit 3: Coordinate Geometry of Three Dimensions-** Equation of a sphere, Intersection of a sphere and a plane, tangent plane, Intersection of two spheres, orthogonality of two spheres, Right circular cone, Right circular cylinder.

**Unit 4: Vector Calculus-** Scalar and vector fields, Differentiation and Integration of vector functions, Directional derivatives, Gradient, Divergence and Curl.Line, Surface and volume Integrals. Green's theorem in a plane, Gauss's and Stoke's theorem (without proof) and their applications

**Unit 5: Statistics and Probability-** Elementary theory of probability, Baye's theorem with simple applications, theoretical probability distributions: Binomial, Poisson and Normal.

**Text/Reference Books**


1: Analytic Solid Geometry, Shanti Narayan (S.Chand).
2: Advanced Engineering Mathematics, Erwin Kreyszig, Wiley 9$^{th}$Edition.
3: Mathematics for Engineers, Chandrika Prasad, Prasad Mudranalaya Allahabad.
4: Advanced Mathematics for Engineers, Chandrika Prasad, Prasad Mudranalaya.
5: Vector Analysis, M.D.Raisinghania (S.Chand).
6: Advanced Engineering Mathematics, Jain and Iyenger, Narosa.
8: Probability,Statistics and Queueing theory:Allen.
9: Statistical Methods (Vol. I & II), N.G.Das, TMH.
10: Schaum's Outline Series for Vector Calculus.
11: Schaum's Outline Series for Matrices.
12: Engineering Mathematics-II by C.B. Gupta and A .K. Malik, New Age International Pvt. Ltd.
13: Higher Engineering Mathematics, B.V.Ramana, Tata McGraw Hill.14: Probability, Statistics and Queueing theory, V.Sundrapandian, PHI.

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## PHY 107:PHYSICS FOR ENGINEERS -II

**Unit I: Special Theory of Relativity:** Postulates of special theory of relativity, Lorentz transformations, Relativity of length, mass and time. Relativistic velocity addition, mass-energy relation. Relativistic Energy and momentum.

**Unit II: Quantum Mechanics:** Compton Effect & quantum nature of light. Schrodinger's Wave Equation: Time dependent and time independent cases. Physical interpretation of wave function and its properties, boundary conditions.
**Applications of Schrodinger's Equation:** Particle in one-dimensional box. Particle in three-dimensional boxes. Degeneracy, Barrier penetration and tunnel effect, Alpha Decay.

**Unit III: Metals and Superconductors: Summerfield's Free electron gas model:** Postulates, Density of energy states, Fermi energy level, Band Theory of solids.
**Superconductivity:** Qualitative study of the phenomenon-meissner's effect-Josephson effect Type I & II super conductor.

**Unit IV: Electro Dynamics:** Scalar and Vector fields, Definitions of gradient Divergence and curl, Maxwell's Equations.

**Unit V: Nuclear Radiations Detector:** - Nuclear radiations detector: Principle of Gas filled detector. Proposnal counter,G M counter,Scintillation counter.

**Text/Reference Books**


1. Concept of Modern Physics, Arthur Beiser, Tata McGraw Hill (Noida).
2. Introduction to Solid State Physics, C. Kittel, Wiley Editions(New Delhi).
3. Introduction to Electrodynamics, John D. Griffith, Tata McGraw Hill(Noida).
4. Electromagnetics, Sadiku, Oxford university Press(New Delhi).
5. QuantumMechanics, Ghatak & Loknathan,Tata McGraw Hill (Noida).

**Unit 1: Introduction and natural resources:** Multidisciplinary nature and public awareness, renewable and nonrenewable resources and associated problems, forest, water, mineral, food, energy and land resources. Introduction to natural resources, conservation of natural resources and human role.

**Unit 2: Ecosystem:** Ecological concepts, concept of ecosystems, types of ecosystems, ecosystem structure and functioning, energy flow, food chains and food webs, ecological pyramids.

**Unit 3: Biodiversity and Conservation:** Definition, genetic species and ecosystem diversity biogeographically, classification of Indian value of biodiversity at national and local levels, India as a mega-diversity nation, treats to biodiversity and endangered and endemic species of India, need for conservation of biodiversity.

**Unit 4: Environmental pollution:** Definition, causes, effect and control of air pollution, water pollution, soil pollution, marine pollution, noise pollution, thermal pollution, electromagnetic pollution, nuclear hazards, human role in prevention of pollution, solid waste management, disaster management, floods, earthquake, cyclone, and landslide

**Firework Safety:** Combustion of firework and pollution (noise, smoke, fireworks fallout and residue pollution), heavy metal toxicity due to fireworks and associated health effects.

**Unit 5: Social Issue and Environment:** Unsuitable to suitable development, urban problem related to energy and water conservation, environment protection act, wild life protection act, forest conservation act, environmental issues, population explosion, and family welfare programme. Environmental and human health HIV, women and child welfare, role of information technology on environment and human health.

**Corruption:** definition and reasons, details of organizations/agencies working against corruption, role of individual against corruption and mode of action.

**Ethics :** Meaning , nature, determinants and objectives of ethics, ethics and its relation to values norms and morals, Indian ethos, Swami Vivekananda and ethics.

**Text/Reference Books**


1. Fundamentals of Environmental Biology, K. C. Agrawal, Nidhi Publishers(Bikaner).
2. Fundamentals of Ecology, E.P. Odum, W.B. Saunders Co. (USA).
3. Fundamentals of Ecology, E. P. Odum, Natraj Publisher (Dehradun).
4. Ecology: Principles and Applications, J. L. Chapman & , M. J. Reiss, Cambridge University Press.
5. Atmospheric pollution, W. Buch , Tata McGraw Hill(TMH)
6. Professional Ethics and Human Values, M. Govindarajan, PHI Learning Private Limited (Delhi).
7. Corruption and Reform in India, Jennifer Bussell , Cambridge University Press.

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# BCSI201: DESIGN THINKING

**Course Objectives:**

- The human centered design process focuses on understanding and empathizing with the users and the context, and then on defining the users' needs in an actionable way.
- After developing a design paradigm, the process engages the first of many cycles of rapid experimentation, prototyping, and testing with users to drive the evolution of a solution that most effectively satisfies user's needs.

**Course Outcomes:**

Understand how teaching and learning occurs in the design process

- Recognize the ethical and social dilemmas and obligations of the practice of design
- Diagnose common adoption barriers in individuals, groups and organizations.
- Develop a design theory from independent and qualitative research and observations.
- Participate in and lead innovation in creative and collaborative settings
- Undertake complex and unstructured problem-solving challenges in unfamiliar domains

**Unit I: Design Thinking**

Foundations of Human Centered Design, Barriers to Innovation and Adoption, Learning by Doing, Understanding Needs in Context.

**Unit II: Design Needs and Interventions**

The Ethics of Design Interventions, Design Needs in Education, Engineering and Health & Society.

**Unit III: Empathy in Design**

Discovering Explicit and Latent Needs, Qualitative Research: Watching and Listening, Point of View & Problem Reframing, Developing Grounded Theory, Design for Usability

**Unit IV: Ideation, Experimentation and Evolution**

Generating and Developing Ideas, Creativity as Teaching and Learning, Prototyping and testing Learning Through Things & Interactions, Express, Test, Cycle

**Unit V: Design Documentation**

Representing Design Knowledge, Diffusion of Innovation, Design as research

**Textbooks:**

1. Design Thinking: Integrating Innovation, Customer Experience, and Brand Value,byThomas Lockwood,Allworth Press, 2010.
2. 101 Design Methods: A Structured Approach for Driving Innovation in Your Organization Book by Vijay Kumar.

**Reference Books:**

1. Design Thinking: Understand – Improve – Applyedited by Hasso Plattner, Christoph Meinel, Larry Leifer, Springer Science & Business Media, 2010

***Latest editions of all the suggested books are recommended.**

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# BCSI202: Web Designing

## UNIT I

### Introduction to the Internet and the World Wide Web

Introduction, History of internet, Internet Design Principles, Internet Protocols - FTP, TCP/IP, SMTP, Telnet, etc., Client Server Communication, Web System architecture

Evolution of the Web, Web architectures, Web clients and servers, Static and Dynamic Web Applications, Front end and back end web development. HTML, CSS, JS, XML; HTTP, secure HTTP, etc; URL, Web Services – SOAP, REST.

## UNIT II

### HTML & CSS

Introduction to Html, Html Document structure, Html Editors, Html element/tag & attributes, Designing simple page - Html tag, Head tag, Body tag; More Html tags - Anchor tag, Image tag, Table tag, List tag, Frame tag, Div tag ; Html forms - Input type, Text area, Select , Button, Images.

Introduction to CSS, Syntax, Selectors, Embedding CSS to Html, Formatting fonts, Text & background colour, Inline styles, External and Internal Style Sheets, Borders & boxing.

## UNIT III

### XML and HTML5, CSS3

Introduction to XML, Difference b/w Html & XML, XML editors, XML Elements & Attributes XML DTD, XML Schema, XML Parser, Document Object Model (DOM), XML DOM.

## UNIT IV

Introduction to HTML5, New features, Local storage, Web Sockets, Server events, Canvas, Audio & Video, Geolocation, Microdata, Drag and Drop. Browser life cycle and browser rendering stages. Service workers.

**UNIT V**

**PHP Server side scripting**

Introduction to PHP, Basic Syntax, Variables, constants and operators, Loops, Arrays and Strings, Environment & environment variables, responding to HTTP requests, Files, Cookies, Sessions, Examples.

**Books:**


1. Practical Web Design for Absolute Beginners, Adrian W. West. Apress 2016
2. Introducing Web Development, Jorg Krause. Apress 2017.
3. HTML & CSS: The Complete Reference, Thomas Powell. McGraw Hill, Fifth Edition, 2010
4. Creating a Website: The Missing Manual, 3rd Edition, Mathew Macdonald. O'Reilly
5. Web Technologies - HTML, JavaScript, PHP, Java, JSP, ASP.NET, XML and Ajax Black, Kogen Learning Systems (Dreamtech Press), 5th Edition 2009.
6. HTML, XHTML & CSS Bible, Brian Pfaffenberger, Steven M.Schafer, Charles White, Bill Karow- Wiley Publishing Inc, 2010
7. HTML5 & CSS3 for the Real World, 2 Edition, Alexis Goldstein, Estelle Weyl, Louis Lazaris. Apress 2015.
8. HTML5 & CSS3 for Dummies, Andy Harris. Wiley 2014.
9. Learning PHP A Gentle Introduction to the Web's Most Popular Language, David Sklar. O'Reilly 2016.
10. Build Your Own Database Driven Web Site Using PHP & MySQL, Kevin Yank. Sitepoint , 4th Edition, 2009.